

Anas TBER

Data Science & AI Engineering Student — Centrale Lyon

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Engineering student at Centrale Lyon with hands-on experience in ML, NLP, LLM applications and large-scale data engineering. Seeking an **end-of-studies internship in Data Science or AI** starting **March 2027**.

Education

- École Centrale de Lyon** — *General Engineering* Lyon · Sep. 2023 – Mar. 2027
- **Courses:** Applied Statistics for Data Science, Machine Learning & Applications, Bayesian Learning & Text Mining, Big Data Technologies, Representation & Manipulation of Structured Data, Computer Vision, Probability & Stochastic Processes, Enterprise Management
- Lycée Henri Loritz** — *CPGE, PCSI / PSI** Nancy · Sep. 2020 – Jun. 2023
- Intensive preparatory classes in Physics, Mathematics & Engineering Sciences for France's top engineering school entrance exams (Grandes Écoles).

Experience

- Data Scientist Intern — BNP Paribas CIB** — *Compliance, Anti-Money Laundering (AML)* Paris · Jan. 2026 – Jul. 2026
- Delivered a **regulator-approved** network-based AML detection & investigation platform on a **10M+ entity / 40M+ transaction** dataset, from raw ingestion to compliance dashboards; built the pipeline in **Polars / PyArrow** lazy mode with watchlist enrichment
 - Designed the detection layer on engineered graph features (centrality, pass-through ratio, flagged-neighbour ratio): **cycle detection** (SCC + DFS) to surface money-mule rings, and an **Isolation Forest** model with **SHAP** explainability ranking suspicious communities with regulator-facing justifications
 - Shipped **LLM-powered** features into the **Streamlit** app: one-click AI risk summaries of a community graph and a multi-turn **investigation chatbot** answering free-form questions on an account and its 3-hop neighbourhood (prompt engineering, context-window management)
 - **Sole owner and maintainer of the repository** — led the refactor (modular architecture, centralized configuration) and wrote the **full technical documentation** enabling handover to the IT team
- Tech & Operations Intern — Gentle Mates** — *Professional Esports* Paris · Apr. 2025 – Nov. 2025
- Built an automated social media analytics dashboard (Streamlit/Python) integrating computer vision (**YOLO**) for sponsor logo detection and NLP/**Regex** for brand mention tracking, improving partnership ROI visibility
 - Contributed to the development of impactful software and AI internal tools for analytics, monitoring, and operations

Selected Personal Projects

- AirKeys — Camera-Based Gesture Instrument** — *Self-Trained ML Pipeline* 2026 · [Live demo](#)
- Built a **hand gesture recognition** system combining **MediaPipe**'s pretrained landmark model with a self-trained **MLP** classifier (**scikit-learn**, benchmarked against **Random Forest**) on a self-collected, hand-labeled dataset, reaching **99.8%** held-out accuracy
 - Enabled **on-the-fly personalization**: users can teach the app new gestures live via a from-scratch **k-nearest-neighbors** classifier, requiring no retraining and running entirely client-side
- arXiv Research Radar — RAG over Recent Papers** — *Personal Project* 2026 · [GitHub repo](#)
- Built an end-to-end **RAG** system answering questions over the latest arXiv papers with per-claim citations: PDF ingestion, **section-aware chunking**, **Sentence-Transformers** embeddings with **FAISS** search, and **Gemini** answers grounded only in retrieved sources (prompt-injection guard)
 - Served via **FastAPI + Streamlit** with cost controls (per-IP rate limiting, atomic SQLite daily cap, relevance threshold that skips the LLM on off-topic questions); index refreshed weekly by a **GitHub Actions** pipeline

Skills

Programming	Python, JavaScript
AI & Data Science	Pandas, NumPy, Scikit-Learn, PyTorch, LightGBM, SHAP, LangChain, Jupyter
LLM & NLP	RAG (embeddings, FAISS), Gemini/Anthropic/OpenAI APIs, Transformers, Prompt engineering,
Data Engineering	Polars, PyArrow, SQL
Web & Viz	Streamlit, Pyvis, Plotly, HTML/CSS
Languages	French (Native), English (Fluent), Arabic(Native), Spanish (Beginner)